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AMB3R: Accurate Feed-forward Metric-scale 3D Reconstruction with Backend

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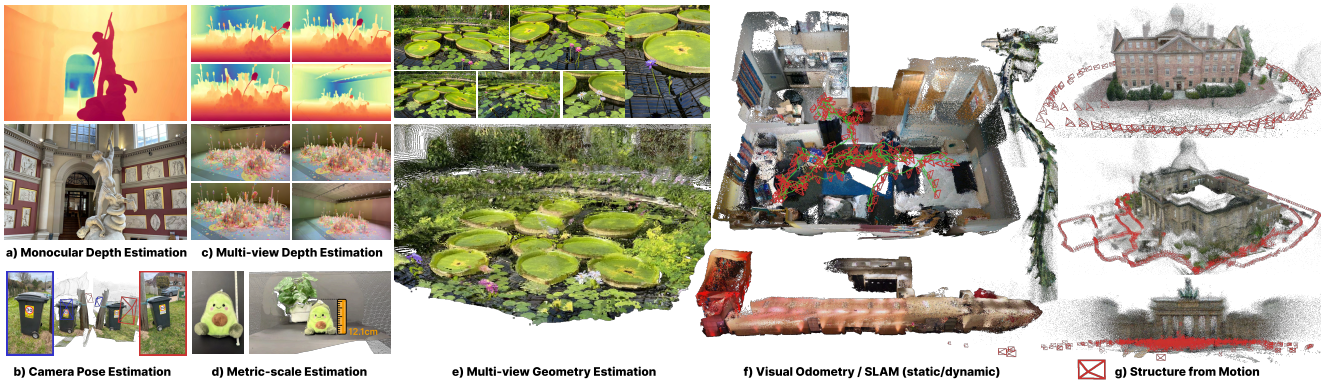


Figure 1. **Overview.** We present AMB3R, a **feed-forward model for metric-scale 3D reconstruction**, which supports camera pose estimation, depth estimation, and 3D reconstruction. We also propose AMB3R-VO and AMB3R-SfM, two **training-free, model-agnostic pipelines** that enable multi-view networks to perform feed-forward Visual Odometry (VO) and Structure from Motion (SfM) for any number of images without the need for test-time optimization. We use in-the-wild images for a)-e), scenes from Co-SLAM [110], TTT3R [15], and KITTI [33] for f), scenes from COLMAP [90], Tanks&Temples [50], and IMC PhotoTourism [44] (all images) for g).

Abstract

We present AMB3R, a multi-view feed-forward model for metric-scale dense 3D reconstruction that addresses diverse 3D vision tasks. The key idea is to employ a sparse, yet compact, volumetric scene representation as our backend, enabling geometric reasoning with spatial compactness. We further introduce AMB3R-VO and AMB3R-SfM, two training-free, model-agnostic pipelines that extend multi-view networks to uncalibrated visual odometry (online) or structure from motion for any number of images without test-time optimization. Compared to prior 3D foundation models, AMB3R achieves state-of-the-art performance in camera pose, depth, metric-scale estimation, and 3D reconstruction, while AMB3R-VO and AMB3R-SfM even surpass optimization-based SLAM and SfM systems with dense reconstruction priors on common benchmarks.

1. Introduction

Pointmaps have emerged as the cornerstone of modern 3D foundation models. By exploiting the one-to-one correspondence between 2D pixels and 3D point coordinates, we can train a neural network to tackle various 3D vision tasks. Yet, this raises a fundamental question: *Is the mapping from 2D pixels to 3D scene points truly one-to-one?*

In practice, this is not the case. Due to visual overlap, multiple pixels often correspond to the same 3D point. This many-to-one mapping, known as **correspondence**, lies at the heart of decades of research in 3D vision. Early sparse reconstruction methods [3, 19] leverage correspondences between interest points to recover camera parameters and sparse geometry. Multi-view stereo (MVS) methods [31] construct cost volumes for depth estimation, which essentially match pixel correspondences within the volume. For dense 3D reconstruction, approaches ranging from traditional methods (e.g., KinectFusion [80]) to modern neural implicit models (e.g., DeepSDF [84], NeRF [75], NeuralRecon [99]) adopt compact 3D representations such as TSDF grids, feature grids, or coordinate-based networks. Despite their diverse forms, these representations share a commonality—**spatial compactness** (i.e., the same 3D coordinate can only have a unique property). This compactness enforces the fusion of multiple corresponding observations of the same scene point into a coherent geometry.

Recent feed-forward models such as Spann3R [109] or VGGT [111] follow the DUSt3R [114] paradigm, formulating the task as per-pixel 2D-to-3D regression. While the pointmap output space implicitly encourages multiple cor-

responding pixels to have the same 3D location regardless of viewpoint, the network itself operates on 2D grids and lacks explicit geometric reasoning or spatial compactness.

In this paper, we propose AMB3R, a multi-view network with a compact scene representation as backend. We use VGGT as the front-end to predict features and geometry, and introduce a lightweight scale head for metric depth. The backend uses sparse voxel representations serialized via space-filling curves and processed by a transformer, enabling efficient reasoning in compact 3D space. The fused features are injected back into the front-end decoder through zero-convolution [127], allowing reuse of pre-trained weights and confidence of the front-end, substantially reducing the training cost ($\sim$ 80 H100 GPU hrs).

To address quadratic complexity and the offline limits of multi-view transformers, we propose AMB3R-VO and AMB3R-SfM: **training-free**, **model-agnostic** pipelines for feed-forward visual odometry and structure-from-motion of any number of images without test-time optimization. AMB3R-VO leverages a strong prior in pointmap models, enabling online mapping with $\mathcal{O}(1)$ per-frame complexity via automatic keyframe selection and active memory management. Built on the same prior and memory design, AMB3R-SfM employs a divide-and-conquer strategy to solve large-scale reconstruction from unordered image collections incrementally. To sum up, our contributions are:

1. **A Compact Backend Scene Representation** for a feed-forward pointmap-based foundation model that enables explicit 3D reasoning with spatial compactness.
2. **Metric-scale Reconstruction** with a light-weight scale head to recover metric depth information.
3. **Uncalibrated Visual Odometry** without the need for fine-tuning or an optimization-based backend. Notably, we show for the first time, feed-forward visual odometry can surpass its optimization-based counterparts.
4. **Feed-forward Structure from Motion** capable of large-scale reconstruction without optimization.
5. **State-of-the-Art Performance** on 7 tasks across 13 datasets, surpassing existing 3D foundation models and only requiring academic-level training resources ($\sim$ 80 H100 GPU hrs) to train the metric-scale head and new backend on top of frozen VGGT [111] weights.
6. **Open Source Release** of code, weights, and an evaluation toolkit to facilitate future research.

2. Related Work

2.1. 3D Scene Representations

A 3D scene can be represented by properties such as SDFs, volume density, radiance, surface normals, or learned features, stored in spatial data structures including voxel grids, point clouds, meshes, or even MLPs. Among these, voxel grids are particularly popular in geometric deep learning as a natural 3D extension of 2D pixels. Since grids grow cubically with scene size, many sparse alternatives have been introduced, including lossless representations (e.g., voxel hashing [81], octrees [41, 73]) and lossy ones (e.g., hash grids [76], triplanes [12, 85], vector–matrix decompositions [13]), many of which rely on a low-rank approximation of the 3D scene. Point clouds are another common spatial data structure as they are readily available from posed RGB-D images. Unlike voxels, point clouds are irregular and sampled near visible surfaces. To process such data, HilbertNet [14] maps points onto a Hilbert curve [1, 36], a fractal space-filling curve that traverses the 3D space while preserving spatial locality, thereby serializing points into a 1D sequence for efficient processing [59, 116, 117, 131].

2.2. Pointmap-based Reconstruction

Pointmaps, or scene coordinates, were introduced by SCoRF [96] as a scene-specific representation for camera localization [5, 7, 8, 88]. DUSt3R [114] and MASt3R [53] extend this idea by training neural networks for two-view pointmap regression in a generalizable manner. Since pointmaps inherently encode both camera information and scene geometry, they provide a unified representation for diverse geometric vision tasks. To scale beyond two views, Spann3R [109] and others [10, 15, 16, 49, 52, 58, 64, 68, 70, 112, 118, 122, 137] adopt an incremental paradigm in a similar spirit to incremental SfM/SLAM [19, 90], whereas VGGT [111] and others [26, 101, 108, 115, 120] follow a global paradigm that jointly processes all images as in global SfM [3, 83]. Beyond dense reconstruction, extensions have been proposed for calibrated settings [42, 48], camera calibration [67], camera localisation [23, 38, 63, 65], monocular geometry estimation [29, 46, 55, 66, 100], dynamic reconstruction [30, 43, 54, 71, 102, 128], novel view synthesis [39, 56, 60, 61, 74, 94, 119, 129], surface reconstruction [135], semantic segmentation [138], spatial reasoning [28, 37, 87], and scientific imaging [126]. Since feed-forward approaches are typically constrained to moderate scene sizes [109] or limited image counts [111], several works [20, 24, 78, 124, 134] also explore optimization-based formulations to scale up reconstruction.

3. Preliminaries: Pointmap-based Front-end

Given a set of input images $\{I_t\}_{t=1}^{T}$, pointmap-based models independently encode each image into visual features:

$$F_t = \text{Encoder}(I_t) \tag{1}$$

Since the pointmap is defined up to a rigid transformation, we fix an anchor image and set its pose as identity. The goal is then to predict a pointmap per image in the anchor coordinate system. This requires a temporal encoding that (*i*) disentangles visual features across images while being gen-

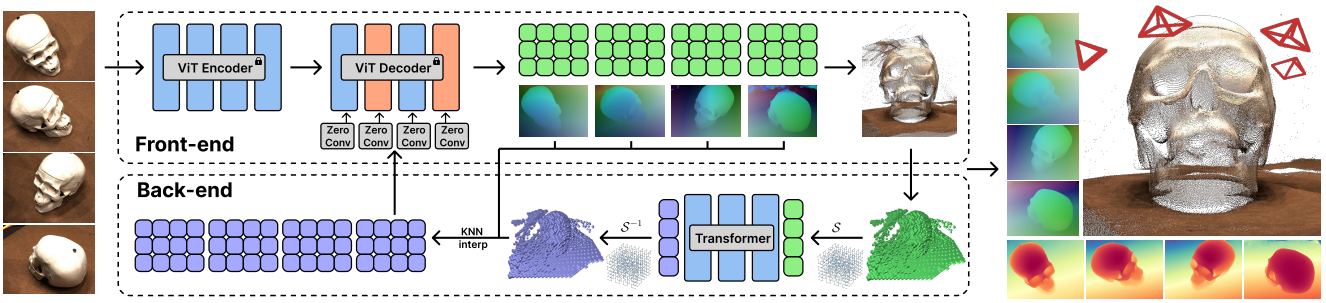


Figure 2. **Overview of AMB3R.** AMB3R consists of a front-end that predicts pointmaps and geometric features, and a back-end that fuses them into sparse voxels, which are serialized into a 1D sequence, processed by a transformer, and unserialized back to 3D. Per-pixel features are obtained via KNN interpolation and injected into the frozen front-end via zero-convolution for final prediction.

eralizable to many images, and (*ii*) identifies the anchor image to provide a consistent reference frame. Existing models can be categorized into three paradigms:

(a) Network-based temporal encoding. DUSt3R [114] employs two intertwined decoders for two images, where the coordinate system is defined by the reference decoder:

$$P_1^{(1)} = \text{Decoder}^1(I_1), \quad P_2^{(1)} = \text{Decoder}^2(I_2). \tag{2}$$

This design choice satisfies condition (*ii*), but inherently limits its scalability to only two images.

(b) Causal temporal encoding. Spann3R [109] proposes an incremental reconstruction framework, where each incoming frame queries accumulated memory features:

$$P_t^{(1)} = \text{Decoder}(M_{1:t-1}, F_t). \tag{3}$$

This formulation naturally satisfies (*i*) and (*ii*) in a causal setting. By sparsifying memory features, Spann3R can process long sequences without exploding GPU memory.

(c) Token-based temporal encoding. VGGT [111] uses two camera tokens e for the anchor and other images with alternating attention as a generalizable temporal encoding:

$$\{P_t^{(1)}\}_{t=1}^T = \text{Decoder}\big(F_1 + e_1, \{F_t + e_2\}_{t=2}^T\big). \tag{4}$$

This enables offline reconstruction with densely connected attention, leading to superior reconstruction quality.

Coordinate normalization. Pointmap prediction from images is defined up to scale. During training, DUSt3R and Spann3R normalize both predictions and ground truth, whereas VGGT normalizes only the ground truth and trains in this canonical space, resulting in more stable training.

Confidence-aware supervision. These models are trained with a confidence loss on the ground-truth geometry:

$$\mathcal{L}_{\text{conf}} = \sum_t \sum_{i \in \mathcal{I}} \left(C_t^i \mathcal{L}_{\text{reg}}(i) - \alpha \log C_t^i \right), \tag{5}$$

where $\mathcal{I}$ denotes valid pixels. Confidence C_t is learned in a self-supervised way and depends on the data distribution and training recipe. Fine-tuning without careful initialization can cause catastrophic forgetting of the learned confidence, degrading reconstruction performance.

4. Method

4.1. Metric-scale Reconstruction

VGGT [111] predicts pointmaps normalized by the median distance across all frames (Sec. 3), suggesting that it implicitly encodes metric cues. Thus, we introduce a lightweight scale head to recover metric scale from its frozen features.

A simple solution would be to regress the global scale difference between the ground truth and the prediction using an ROE solver [113]. Since this global scale depends on ALL frames, this approach requires aggregating intermediate decoder features of all frames. In practice, though, we find this approach difficult to train and prone to overfitting, as the global scale difference can be sensitive to different frame combinations and permutations.

Instead, we propose to regress the **metric log depth** corresponding to the pixel with the median predicted depth for each frame independently. This captures an intrinsic per-frame property recoverable from individual encoder features, decoupling scale prediction from the model's global prediction and making the task substantially easier. We further concatenate decoder depth features with encoder features to improve scale estimation. At inference, we can use per-frame median predicted depth and metric depth to estimate per-frame scales and use their median to robustly align the global reconstruction to metric space.

4.2. Backend

Given pointmap predictions $\{P_t^{(1)}\}_{t=1}^T$ and geometric features $\{G_t\}_{t=1}^T$ extracted from concatenated encoder and decoder features via a two-layer MLP, we align their resolution and construct a sparse voxel grid $\mathcal{V}$. Voxel features $\{H_i\}_{i=1}^N$ are obtained by averaging features within each voxel $\mathcal{V}_i$:

$$H_i = \frac{1}{|\mathcal{P}_i|} \sum_{(t,\mathbf{u}) \in \mathcal{P}_i} G_t[\mathbf{u}], \tag{6}$$

where $\mathcal{P}_i = \{(t, \mathbf{u}) \mid P_t^{(1)}[\mathbf{u}] \in \mathcal{V}_i\}$, $\mathbf{u}$ denotes the pixel coordinates. We set the voxel size to 0.01 in normalized space, naturally enabling adaptive voxel resolution w.r.t.

scene scale. The sparse voxel grid is serialized into 1D feature sequences via space-filling curves, processed by a transformer, and deserialized back to voxel space:

$$\{\hat{H}_i\}_{i=1}^{N} = (\mathcal{S}^{-1} \circ f_\theta \circ \mathcal{S})\left(\{H_i\}_{i=1}^{N}\right), \tag{7}$$

where $\mathcal{S}$ denotes the serialization and f_θ is implemented via Point Transformer v3 [117]. We then obtain per-point features using KNN interpolation:

$$\tilde{G}_t[\mathbf{u}] = \mathrm{KNN}\big(P_t^{(1)}[\mathbf{u}], \{\hat{H}\}_{i=1}^{N}\big). \tag{8}$$

These features are fused back to each decoder layer using zero convolution, allowing the model to reuse the attention and confidence functions learned by the front-end model while substantially reducing training cost. See Fig. 2.

4.3. Training

We freeze the front-end model (i.e., VGGT [111]) and train only the backend using the same losses as VGGT for depth, pointmap, and camera pose (excluding tracking):

$$\mathcal{L} = \mathcal{L}_{\text{depth}} + \mathcal{L}_{\text{pointmap}} + \mathcal{L}_{\text{camera}}. \tag{9}$$

Since our training data, preprocessing scripts, and losses differ from VGGT, the canonical scale learned by minimizing $\mathcal{L}$ may not align with that of VGGT. Direct supervision on normalized data would therefore force the model to waste capacity compensating for this mismatch. Instead, we align the predicted geometry with the normalized ground truth using the ROE [113] solver before supervision.

Specifically, for pointmap, we estimate a single scale per sequence, while for depth, we relax this constraint and estimate an independent scale per image. As the front-end is trained to produce geometry in a consistent scale and remains frozen, this allows the backend to focus on refining fine structural details without compromising global consistency. The backend is trained for 40 epochs on a mixture of 11 datasets (2000 samples per epoch, 80K total) with 5–16 frames per sample, requiring ∼50 H100 GPU hours.

4.4. Uncalibrated Visual Odometry

Multi-view transformers (type-(c) methods) typically exhibit quadratic computational complexity, making them impractical for online VO/SLAM. Existing methods (e.g., VGGT-SLAM [69]) address this by building overlapping submaps sequentially, estimating the relative transformation and scale difference using the Kabsch–Umeyama algorithm [45, 105]. Since such alignment introduces non-negligible drift, they rely heavily on optimization-based backends [69]. However, this overlooks a powerful prior in pointmap foundation models: predictions are expressed in the reference frame's coordinate system, up to an unknown scale. This observation eliminates the need for explicit transformation estimation for coordinate alignment.

Building on this prior, we propose AMB3R-VO, a training-free, model-agnostic visual odometry pipeline for any number of images with constant per-frame complexity via a hybrid memory design. Our key idea is to use **keyframes as memory**. Rewriting Eq. 4 as

$$\{P_t^{(1)}\}_{t=1}^{T-1}, P_T^{(1)} = \mathrm{Decoder}\big(\mathcal{M}_{1:T-1}, F_T + e_2\big), \tag{10}$$

where $\mathcal{M}_{1:T-1} = \big\{F_1 + e_1, \{F_t + e_2\}_{t=2}^{T-1}\big\}$. Compared to Eq. 3 under the streaming setting, type-(c) methods can be viewed as a special memory network that also updates the previous predictions with bi-directional attention.

To prevent memory growth, we adopt a hybrid memory strategy: a small set of carefully sampled keyframes is maintained as the *active memory* for the forward pass of the base network, while a separate *global memory* stores explicit geometry representations for coordinate alignment and fusion. This design limits the size of $\mathcal{M}$ by a fixed number of active keyframes, ensuring constant mapping complexity while preserving global consistency.

Keyframe selection. We can measure the pose distance as in VGGT [111] and our training frame sampling strategy:

$$D_{i,j} = \arccos\left(\frac{\mathrm{Tr}(R_j R_i^T) - 1}{2}\right) + \lambda\|\tau_i' - \tau_j'\|_2, \tag{11}$$

where R and τ' are rotation and normalized translation vector. We set a minimal pose distance with respect to the poses of existing keyframes $\eta_d = 0.15$ and iteratively add the frame with the highest confidence within each mapping window to the active memory as a new keyframe. The active memory serves as context for mapping the next window of $N_w = 8$ frames. Since predictions are up to an unknown scale, we estimate the new window's scale relative to the global map using their shared keyframes k in window w:

$$s^w = \mathrm{ROE}(P_k^{(1)}, P_k^{(1),w}) \tag{12}$$

Global memory. We represent global memory as explicit global map geometry and update it with each mapping window result in a running average manner:

$$P_k \leftarrow (C_k P_k + C_k^w s^w P_k^w)/(C_k + C_k^w), \tag{13}$$
$$m \leftarrow (C_k m + C_k^w m^w / s^w)/(C_k + C_k^w), \tag{14}$$
$$\tau_k \leftarrow (C_k \tau_k + C_k^w s^w \tau_k^w)/(C_k + C_k^w), \tag{15}$$
$$q_k \leftarrow \mathrm{slerp}(q_k, q_k^w, C_k^w/(C_k + C_k^w)), \tag{16}$$
$$C_k \leftarrow C_k + C_k^w. \tag{17}$$

where q_k is the quaternion vector of the rotation matrix. C_k is the predicted confidence. m is the metric-scale factor.

Active memory management. Once the number of keyframes in active memory reaches $N_{max} = 10$, we resample $N_{min} = 7$ keyframes from all historical keyframes. The most recent keyframe is always preserved. We then

select N_{topk} keyframes with the smallest pose distance to the latest keyframe. To encourage the model to use earlier keyframes to form a loop, we introduce a backward search window $\eta_b = 0.4$ and iteratively add the earliest keyframe within this window, resulting in N_{old} keyframes. Finally, a frame with the minimal summed pose distance to the current set is chosen as the first frame. If the maximum pose distance among active keyframes exceeds $\eta_{max} = 1.2$, we resample active memory to ensure the model only covers a reasonable scene region within the training distribution.

Coordinate alignment. After keyframe resampling, the first keyframe in active memory may not be the first frame, resulting in a different coordinate system between our global memory and local prediction. Instead of jointly estimating scale and relative transformation from the local map to the global map, we can first map our global map into the local map:

$$P_k^{(k_0)} = T_{k_0}^{-1} P_k^{(1)}, \tag{18}$$

where $T_{k_0} = [R_{k_0}, \tau_{k_0}]$. We can then estimate the scale between $P_k^{(k_0)}$ and $P_k^{(k_0),w}$. After scaling the camera pose, we can obtain the relative pose of each corresponding keyframe and use their weighted average to map the local map back to the global frame of reference and fuse them as in Eq. 13-17, thus avoiding explicit Kabsch–Umeyama alignment.

Robust estimation. To reduce inference cost without sacrificing performance, we run the backend only when the confidence of the front-end prediction falls below a predefined threshold. Based on both confidence and prediction consistency, we choose either prediction or blend them, ensuring robust performance in the sequential system.

4.5. Structure from Motion

Building on the same prior and memory design as in AMB3R-VO, we propose AMB3R-SfM, a training-free, model-agnostic structure-from-motion pipeline for any number of images without test time optimization. To handle the unordered image collection, we propose a divide-and-conquer strategy that partitions images into small clusters and solves them following the incremental SfM paradigm.

Image clustering. For each given image I_t, we extract its feature descriptor $\bar{F}_t \in \mathbb{R}^C$, apply feature whitening, and construct the feature distance matrix $D^F \in \mathbb{R}^{T \times T}$. Farthest Point Sampling (FPS) with iterative splitting and merging is then applied to construct image clusters, each containing between $N_{c_{min}}$ and $N_{c_{max}}$ images.

Coarse registration. We initialize the map using the cluster with highest confidence and maintain a global keyframe list with $\eta_d = 0.2$. After initialization, the map is incrementally built by: 1) identifying the top-k unmapped clusters with the smallest feature distance to existing keyframes, 2) mapping with existing keyframes, and 3) updating global map using the cluster with highest confidence. When the number of global keyframes exceeds $N_{k_{max}}$, we partition the keyframe list into sub-clusters based on pose distance. Subsequent clusters are then mapped against the top-k smaller sub-clusters to improve registration accuracy. For each frame, only the prediction from the highest-confidence mapping is kept; low-confidence frames are marked as unregistered and mapped again after registration finishes. Map update and coordinate alignment follow Sec. 4.4.

Global mapping. After coarse registration, we perform a two-stage mapping as refinement. First, we refine all keyframes: starting from the first, we find its top-k closest keyframes within $\eta_r = 1.5$ and perform mapping. We then apply a confidence-prioritized breadth-first search (BFS), adding keyframes in the current mapping window to a priority queue sorted by confidence, thereby traversing the entire keyframe graph, and mapping each keyframe with its closest top-k keyframes. Finally, each non-keyframe is refined by mapping with the top-k closest frames.

Method	NYUv2 [79]		KITTI [33]		ETH3D [91]		ScanNet [18]		DIODE [107]	
	Rel	$\delta_{1.25}$	Rel	$\delta_{1.25}$	Rel	$\delta_{1.25}$	Rel	$\delta_{1.25}$	Rel	$\delta_{1.25}$
Omnidata [25]	7.4	94.5	14.9	83.5	16.6	77.8	7.5	93.6	33.9	74.2
DepthAny v2 [121]	4.3	97.9	8.0	94.4	5.7	98.3	(4.2)	(97.9)	21.6	75.2
Marigold [47]	5.5	96.4	9.9	91.6	6.5	96.0	6.4	95.1	30.8	77.3
Diffusion-E2E[32]	5.2	96.6	9.6	91.9	6.2	95.9	5.8	96.2	30.2	77.9
MoGe [113]	3.6	98.0	5.5	97.6	3.3	98.8	3.5	98.3	22.4	82.3
VGGT [111]	3.6	98.0	8.8	92.7	3.8	97.9	(2.7)	(98.8)	26.9	79.1
AMB3R	3.0	98.9	7.3	95.4	3.2	98.8	(2.7)	(98.9)	24.7	80.7

Table 1. **Monocular depth estimation.** Our method achieves competitive zero-shot monocular depth performance.

	DUSt3R [114]	MASt3R [53]	Fast3R [120]	VGGT [111]	VGGT [111]	AMB3R
Re10K [133]	67.7	76.4	72.7	85.3	81.8	**86.3**

Table 2. **Camera Pose Estimation.** We evaluate AUC@30 by randomly sampling 10 frames from the entire sequence.

5. Experiments

5.1. Setup

Benchmarks. We evaluate across 7 tasks on 13 datasets to show the effectiveness and versatility of our approach:

- **Camera Pose Estimation.** We evaluate against SOTA pointmap-based methods on RealEstate10K [133].
- **Monocular Depth Estimation.** We follow prior works [47] and show zero-shot generalization of our model on five benchmarks: NYUv2 [79], KITTI [33], ETH3D [91], ScanNet [18], and DIODE [107].
- **Multi-View (Metric) Depth Estimation.** We compare against various (metric) depth estimation methods on RMVDB [93], using KITTI [33], ETH3D [91], ScanNet [18], DTU [2], and Tanks & Temples [50] datasets.
- **3D Reconstruction.** We benchmark against pointmap-based models on ETH3D [91] and DTU [2] (RMVDB splits), as well as 7Scenes [96] (Spann3R [109] splits). Note that Replica [97] is within DUSt3R [114] and VGGT [111] training data, and we remove NRGBD [4] due to the potential overlap with synthetic training data.

Approach	KITTI		ScanNet		ETH3D		DTU		T&T		Average	
	rel ↓	$\delta_{1.03}$ ↑	rel ↓	$\delta_{1.03}$ ↑	rel ↓	$\delta_{1.03}$ ↑	rel ↓	$\delta_{1.03}$ ↑	rel ↓	$\delta_{1.03}$ ↑	rel ↓	$\delta_{1.03}$ ↑
a) Classic approaches												
COLMAP [90]	12.0	58.2	14.6	34.2	16.4	55.1	0.7	96.5	2.7	95.0	9.3	67.8
b) Single-view depth												
UniDepthV2 [86]	4.0	55.3	(2.1)	(82.6)	3.7	66.2	3.2	72.3	3.6	68.4	3.3	68.9
DepthAny V2 [121]	6.6	38.6	4.0	58.6	4.7	56.5	2.6	74.7	4.5	57.5	4.8	54.1
c) Depth from frames and poses (w/ per-image range)												
Vis-MVSNet [125]	9.5	55.4	8.9	33.5	10.8	43.3	(1.8)	(87.4)	4.1	87.2	7.0	61.4
MVSFormer++ [11]	4.4	65.7	7.9	39.4	7.8	50.4	(0.9)	(95.3)	3.2	88.1	4.8	67.8
d) Depth from frames and poses (w/o per-image range)												
Robust MVD [93]	7.1	41.9	7.4	38.4	9.0	42.6	2.7	82.0	5.0	75.1	6.3	56.0
MVSA [40]	3.2	68.8	3.7	62.9	3.2	68.0	1.3	95.0	2.1	90.5	2.7	77.0
e) Depth from frames (w/o poses)												
DeMoN [106]	15.5	15.2	12.0	21.0	17.4	15.4	21.8	16.6	13.0	23.2	16.0	18.3
DUSt3R [114]	5.4	49.5	(3.1)	(71.8)	3.0	76.0	3.9	68.6	3.3	75.1	3.7	68.2
Spann3R [109]	7.9	36.2	(3.3)	(67.1)	5.7	58.6	3.5	65.2	4.7	58.5	5.0	57.1
Pow3R [42]	5.7	45.7	(3.2)	(68.8)	3.0	74.7	3.0	74.3	3.3	76.6	3.6	68.0
MUSt3R [10]	4.5	55.0	(4.0)	(59.8)	2.5	80.3	4.6	55.4	(2.6)	(80.4)	3.7	66.2
VGGT [111]	4.5	59.6	(2.3)	(80.8)	1.8	86.3	0.9	95.6	2.4	84.1	2.4	81.3
$\pi3^{\ddagger}$ [115]	2.8	72.9	(2.0)	(83.6)	1.3	92.4	1.3	91.8	1.8	87.3	1.8	85.6
MapAny‡ [48]	4.0	59.4	4.0	60.5	2.8	73.2	3.9	63.7	3.3	73.0	3.6	66.0
AMB3R	2.8	74.4	(1.9)	(85.8)	1.4	90.9	0.9	95.1	1.7	90.2	1.7	87.3

Table 3. **Multi-view depth estimation** on RMVDB [93]. ‡ means concurrent works, and () indicate training dataset. Please refer to supplement for full table.

- **Video Depth Estimation.** We evaluate on Sintel [9], BONN [82], and KITTI [33] without finetuning.
- **VO/SLAM.** Our model is compared against traditional sparse/dense and uncalibrated VO/SLAM methods on static and dynamic scenes from TUM [98] and ETH3D SLAM [92] and pseudo groundtruth on 7scenes [95].
- **SfM.** We compare against SOTA optimization-based SfM on ETH3D [91] and Tanks&Temples [50] datasets.

Metrics. For depth estimation, we report absolute relative error (rel) and inlier ratios of 3% ($\delta_{1.03}$) and 25% ($\delta_{1.25}$). For 3D reconstruction, we evaluate rel ratio as well as accuracy and completeness, which measure distances between prediction and ground-truth. We remove ICP alignment in Spann3R [109] evaluation as pointmap has exact correspondence with ground-truth, in contrast to prior works [110, 136] that require sampling points from their mesh. We use median-scale alignment for all depth & 3D reconstruction tasks except metric and monocular depth estimation. For all VO/SLAM tasks, we evaluate the trajectory using ATE RMSE with the EVO package [34]. All distance metrics are reported with the same unit [cm].

5.2. Monocular depth estimation

We compared our zero-shot monocular depth estimation performance in Tab. 1 with VGGT [111] and various representative works. Our model shows consistent improvement over VGGT and SOTA performance on NYUv2 [79] and ETH3D [91], outperforming models specifically trained for monocular depth estimation on those datasets.

5.3. Camera pose estimation.

Following [111, 114], we evaluate camera pose estimation (see Tab. 2) by randomly sampling 10 frames per scene and evaluating AUC@30. For VGGT, we report their official results (gray) (which include some non-accessible data), and our run of VGGT on datasets available to us. AMB3R outperforms VGGT in both cases with our performance gain resulting purely from the improved geometric features.

Method	KITTI [33]			ScanNet [18]			ETH3D [91]			DTU [2]			T&T [50]		
	rel	$\delta_{1.03}$	$\delta_{1.25}$	rel	$\delta_{1.03}$	$\delta_{1.25}$	rel	$\delta_{1.03}$	$\delta_{1.25}$	rel	$\delta_{1.03}$	$\delta_{1.25}$	rel	$\delta_{1.03}$	$\delta_{1.25}$
MASt3R [53]	56.8	0.1	8.4	(10.5)	(27.0)	(91.0)	40.4	10.3	26.7	129.0	1.7	8.1	63.7	0.0	1.5
MUSt3R [10]	82.4	0.0	0.0	(40.2)	(0.7)	(6.2)	63.3	1.0	3.9	46.8	2.6	25.3	77.0	0.0	0.0
Spann3R [109]	16.4	18.9	76.5	(33.1)	(3.3)	(23.0)	25.1	18.6	65.3	67.8	8.7	42.7	23.3	17.6	57.4
CUT3R [112]	28.6	4.1	29.5	(9.9)	(19.2)	**(96.3)**	29.4	6.4	40.7	138.5	0.7	2.3	46.7	0.1	1.9
MapAny‡ [48]	12.8	16.1	78.2	31.9	7.7	41.2	15.4	26.6	83.1	718.5	0.0	0.2	24.6	15.3	41.5
AMB3R (dec. sc.)	4.5	56.5	97.8	**(9.3)**	(41.4)	(91.7)	12.2	42.1	83.9	96.2	25.6	59.3	9.4	42.6	83.8
AMB3R (enc. z)	8.2	27.9	95.6	(9.3)	(26.9)	(95.2)	8.5	48.3	90.4	240.9	30.3	60.9	6.3	49.9	95.7

Table 4. **Multi-view metric-scale depth estimation** on RMVDB [93]. ‡: concurrent work.

	Method	ETH3D			DTU			7-Scenes		
		rel↓	Acc↓	Cp↓	rel↓	Acc↓	Cp↓	rel↓	Acc↓	Cp↓
Onl.	Spann3R [109]	24.96	47.63	31.87	4.96	1.89	0.64	8.58	4.81	5.48
	MUSt3R [10]	19.91	45.19	82.72	5.26	1.68	1.09	11.57	8.94	14.97
	CUT3R [112]	18.83	38.90	22.93	9.11	3.75	0.96	6.32	2.88	3.26
Offl.	VGGT [111]	6.02	12.81	11.89	0.83	0.22	0.08	5.51	2.32	3.51
	$\pi3^{\ddagger}$ [115]	5.82	10.54	10.42	1.57	0.50	0.29	5.92	2.60	3.42
	MapAny‡ [48]	11.20	21.87	19.76	10.37	4.35	1.54	7.61	3.48	3.33
	AMB3R	4.64	9.98	9.69	0.81	0.22	0.08	4.70	1.75	2.22

Table 5. **Multi-view 3D Reconstruction [cm].** For ETH3D and DTU, we use image tuples from RMVDB [93]. For 7 scenes, we use frames as in Spann3R with improved GT poses (See Tab. 7).

5.4. Multi-view (metric) depth estimation

We conduct a comprehensive evaluation on RMVDB [93] in Tab. 3, where our model establishes a new SOTA on RMVDB, outperforming concurrent pointmap-based models ($\pi3$ [115] and MapAnything [48]) while requiring one order of magnitude less training resources. Our model also outperforms the SOTA depth estimation model MVSA [40], which requires ground-truth poses and intrinsics.

For metric depth evaluation (Tab. 4), we report results under both 3% and 25% inlier thresholds, as 3% is overly strict for metric-depth methods. Our approach achieves the best overall performance under both metrics. Compared to directly regressing scale-differences from decoder features, our log-depth regression yields overall better results. On DTU, larger absolute errors occur in scenes with toy buildings placed on white tables with dark backgrounds, which some models mistakenly interpret as real structures.

5.5. 3D reconstruction

We evaluate 3D reconstruction in Tab. 5 and show some qualitative examples in Fig. 3. For object-level reconstruction (DTU), both our model and VGGT achieve millimeter accuracy with less than 1% error. For outdoor (ETH3D) and indoor (7-scenes) scenes, we achieve SOTA performance compared to concurrent work $\pi3$ [115].

5.6. Dynamic reconstruction

We evaluate video depth estimation in Tab. 6. To align with other depth estimation tasks for static scenes, we additionally report an inlier ratio of 3% for a reality check. Despite



Figure 3. **Qualitative showcase** of generalization to in-the-wild images such as Longmen Grottoes [132].

Method	Sintel			Bonn			KITTI		
	rel↓	$\delta_{1.03}$↑	$\delta_{1.25}$↑	rel↓	$\delta_{1.03}$↑	$\delta_{1.25}$↑	rel↓	$\delta_{1.03}$↑	$\delta_{1.25}$↑
Spann3R [109]	55.2	7.3	46.3	5.5	40.9	97.8	24.7	10.4	64.4
CUT3R [112]	58.4	9.6	46.6	5.1	47.4	98.0	13.6	18.0	81.0
VGGT [111]	28.0	13.6	61.8	3.6	63.2	98.4	8.2	40.4	90.9
π^3 [115]	31.9	13.3	66.8	3.0	72.0	98.8	3.6	59.5	98.8
AMB3R	23.6	14.0	62.7	3.3	66.0	98.5	4.1	53.8	98.3

Table 6. **Video depth estimation on Sintel, Bonn, and KITTI**. We evaluate all methods on all scenes from Sintel and Bonn, rather than selecting 14 and 4 scenes as in prior works. In contrast to Pi3 [115], we did not do specific finetuning on dynamic datasets.

		chess	fire	head	office	pump	redkit	stair	avg
	Pseudo GT [80]	3.5	2.5	**1.4**	9.4	15.1	4.9	**2.9**	5.7
	AMB3R (ALL)	**1.8**	**1.8**	2.1	**4.7**	**2.4**	**1.6**	3.7	**2.1**
PSNR	COLMAP (GT) [6]	25.3	25.7	22.6	25.8	25.0	21.9	24.7	24.4
PSNR	Pseudo GT [80]	21.2	22.1	21.2	22.6	22.6	19.9	23.1	21.8
PSNR	AMB3R	20.6	22.3	20.2	22.0	23.6	21.8	22.2	21.8

Table 7. **VO: ATE RMSE [cm] on 7 scenes**. We show our online model surpasses the pseudo GT prior works used [78, 109] on ATE and achieve on-par novel view synthesis PSNR. COLMAP [6] GT is obtained via all sequences on each scene.

AMB3R not being specifically trained for dynamic scene reconstruction, it still shows consistent improvement over VGGT [111] and competitive reconstruction results compared to concurrent work π3[115], which includes dynamic scenes at training. This lays a foundation for making a VO system possible in dynamic environments.

5.7. Uncalibrated Visual Odometry/SLAM

We present VO/SLAM results in Tab. 7- 11. Our model, Spann3R [109], and MUSt3R [10], together establish a new class of *uncalibrated visual odometry* methods with unknown camera intrinsics. Both MUSt3R and our approach outperform classic optimization-based VO, with our model further reducing the tracking error of previous SOTA from 7.1cm to 3.2cm on TUM and from 11.2cm to 2.6cm on ETH3D. Our method even surpasses pseudo GT used in prior work [78, 109] on 7-Scenes (Tab. 7), where we validate ground-truth reliability through novel-view synthesis.

We further compare against state-of-the-art SLAM systems in Tab. 9, where baseline methods are allowed to use loop closure and global bundle adjustment as post-processing after tracking the last frame. The impact of these post-processing steps can be seen by comparing Tab. 8 and Tab. 9. Following prior work [69, 78], we evaluate on raw keyframe poses for fair comparison. Our method shows for

		fr1								fr2		fr3	
		360	desk	desk2	plant	room	rpy	teddy	xyz	xyz	desk	long	AVG
S	ORB-SLAM3 [77]	X	2.0	X	11.8	X	5.6	X	1.0	0.5	1.3	1.7	X
S	DSO [27]	X	27.2	66.0	6.0	58.6	X	X	3.8	0.3	2.2	9.9	X
S	DPVO [62]	13.1	9.4	6.5	3.0	39.8	3.5	6.2	1.3	0.5	3.5	5.5	8.4
D	TANDEM [51]	X	4.3	33.7	X	X	4.9	43.1	2.4	0.3	2.0	8.3	X
D	MonoGS [72]	14.2	6.3	74.0	9.3	64.9	3.4	35.6	1.6	4.5	133.1	3.3	31.8
D	DeepFactors [17]	17.9	15.9	20.2	31.9	38.3	3.8	56.0	5.9	8.4	26.3	49.0	24.9
D	DepthCov [21]	12.8	5.6	4.8	26.1	25.7	5.2	47.5	5.6	1.2	15.9	68.8	19.9
D	DROID-VO [104]	15.7	5.2	11.1	6.0	33.4	3.2	19.1	5.6	10.7	7.9	7.3	11.4
D	COMO [22]	12.9	4.9	9.5	13.8	27.0	4.8	24.5	4.0	0.7	6.3	10.5	10.8
D	GlORIE-VO[123]	13.1	4.0	8.6	4.1	32.7	2.9	14.5	1.2	0.2	16.1	4.8	9.3
U	Spann3R [109]	20.7	16.1	28.3	57.4	84.8	6.1	92.4	2.1	4.4	20.7	193.9	47.9
U	MUSt3R [10]	8.9	5.1	7.1	5.4	13.4	5.2	6.9	2.7	1.7	15.6	4.3	7.1
U	MUSt3R* [10]	7.8	4.0	4.6	4.0	9.9	4.3	4.2	1.3	1.2	15.3	4.3	5.5
U	AMB3R (ALL)	4.6	1.9	2.8	2.9	5.8	2.3	3.7	1.1	2.1	3.4	5.0	3.2
U	AMB3R (KF)	3.9	1.7	2.3	2.7	5.5	2.2	2.8	0.8	0.6	3.0	4.0	2.7

Table 8. **VO: ATE RMSE [cm] on TUM RGB.** We compare with Sparse (S) versus dense (D) versus dense unconstrained (U) methods. The baseline methods are re-run without Loop Closure and global bundle adjustment. We compare with MUSt3R [10] on all frames while keeping keyframe predictions as a reference.

		360	desk	desk2	floor	plant	room	rpy	teddy	xyz	avg
Calibrated	ORB-SLAM3 [77]	X	1.7	21.0	X	3.4	X	X	X	0.9	-
Calibrated	DeepV2D [103]	24.3	16.6	37.9	165.3	20.3	24.6	10.5	31.6	6.4	37.5
Calibrated	DeepFactors [17]	15.9	17.0	25.3	16.9	30.5	36.4	4.3	60.1	3.5	23.3
Calibrated	DPV-SLAM [62]	11.2	1.8	2.9	5.7	2.1	33.0	3.0	8.4	1.0	7.6
Calibrated	DPV-SLAM++ [62]	13.2	1.8	2.9	5.0	2.2	9.6	3.2	9.8	1.0	5.4
Calibrated	GO-SLAM [130]	8.9	1.6	2.8	2.5	2.6	5.2	1.9	4.8	1.0	3.5
Calibrated	DROID-SLAM [104]	11.1	1.8	4.2	2.1	1.6	4.9	2.6	4.8	1.2	3.8
Calibrated	MASt3R-SLAM [78]	4.9	1.6	2.4	2.5	2.0	6.1	2.7	4.1	0.9	3.0
Uncalibrated	DROID-SLAM[104]	20.2	3.2	9.1	6.4	4.5	91.8	5.6	4.5	1.2	15.8
Uncalibrated	MASt3R-SLAM [78]	7.0	3.5	5.5	5.6	3.5	11.8	4.1	11.4	2.0	6.0
Uncalibrated	VGGT-SLAM [69]	7.1	2.5	4.0	14.1	2.3	10.2	3.0	3.4	1.4	5.3
Uncalibrated	AMB3R (ALL)	4.6	1.9	2.8	3.2	2.9	5.8	2.3	3.7	1.1	3.1
Uncalibrated	AMB3R (KF)	3.9	1.7	2.3	2.7	2.7	5.5	2.2	2.8	0.8	2.7

Table 9. **SLAM: ATE RMSE [cm] on TUM RGB.** We use image stride of 2 and keyframe poses evaluation as in MASt3R-SLAM [78] with all poses evaluation shown as a reference. Note that we did not do any post-processing on trajectories

	cables1	camshake1	einstein1	plant1	plant2	sofa1	table3	table7	avg
Spann3R [109]	33.2	5.1	30.9	4.1	5.7	17.1	19.3	18.9	16.8
MUSt3R [10]	20.7	5.6	15.4	2.3	2.7	15.8	17.6	9.5	11.2
MUSt3R* [10]	20.7	5.3	11.2	1.8	2.7	15.5	17.3	5.5	10.0
AMB3R (ALL)	**2.8**	**2.4**	**2.3**	**1.5**	**1.6**	**3.6**	**3.7**	**2.8**	**2.6**
AMB3R (KF)	2.3	2.1	1.9	0.5	0.9	3.1	2.6	2.4	2.0

Table 10. **VO: ATE RMSE [cm] on ETH3D SLAM [92].** We use the same scene sets and image stride of 2 as in MUSt3R [10]

	sit_xyz	desk_p	walk_rpy	walk_s	walk_h	sit_s	sit_h	walk_x	sit_r	avg
MUSt3R [10]	3.5	3.3	6.7	2.2	5.5	1.5	5.3	7.8	4.9	4.5
MUSt3R* [10]	2.4	2.2	5.0	1.9	3.9	1.3	4.6	5.4	4.0	3.4
MegaSaM [57]	1.1	OOM	3.3	0.6	1.8	0.6	2.5	1.4	2.5	-
AMB3R	**1.3**	**2.1**	**4.2**	**0.9**	**1.9**	**0.8**	**2.2**	**2.0**	**2.0**	**1.9**

Table 11. **VO: ATE RMSE on TUM Dynamic.** MUSt3R* uses re-rendering with trajectory smoothness. MegaSaM uses keyframe & global BA at the end followed by a trajectory smoother.

Scenes	COLMAP [90]		ACE-Zero [8]		VGGSfM [111]		DF-SfM [35]		MASt3R-SfM		AMB3R (SfM)	
	RRA@5	RTA@5	RRA@5	RTA@5	RRA@5	RTA@5	RRA@5	RTA@5	RRA@5	RTA@5	RRA@5	RTA@5
courtyard	56.3	60.0	4.0	1.9	50.5	51.2	80.7	74.8	89.8	64.4	100.0	96.5
delivery area	34.0	28.1	27.4	1.9	22.0	19.6	82.5	82.0	83.1	81.8	91.0	76.6
electro	53.3	48.5	16.9	7.9	79.9	58.6	82.8	81.2	100.0	95.5	95.6	81.2
facade	92.2	90.0	74.5	64.1	57.5	48.7	80.9	82.6	74.3	75.3	100.0	95.4
kicker	87.3	86.2	26.2	16.8	100.0	97.8	93.5	91.0	100.0	100.0	100.0	99.2
meadow	0.9	0.9	3.8	0.9	100.0	96.2	56.2	58.1	58.1	58.1	100.0	95.2
office	36.9	32.3	0.9	0.0	64.9	42.1	71.1	54.5	100.0	98.5	100.0	53.9
pipes	30.8	28.6	9.9	1.1	100.0	97.8	72.5	61.5	100.0	100.0	100.0	87.9
playground	17.2	18.1	3.8	2.6	37.3	40.8	70.5	70.1	100.0	93.6	98.7	62.2
relief	16.8	16.8	16.8	17.0	59.6	57.9	32.9	32.9	34.2	40.2	100.0	90.1
relief 2	11.8	11.8	7.3	5.6	69.9	70.3	40.9	39.1	57.4	76.1	100.0	75.7
terrace	100.0	100.0	5.5	2.0	38.7	29.6	100.0	99.6	100.0	100.0	100.0	97.2
terrains	100.0	99.5	15.8	4.5	70.4	54.9	100.0	91.9	58.2	52.5	91.6	53.8
Average	49.0	47.8	16.4	9.7	65.4	58.9	74.2	70.7	81.2	79.7	98.2	81.9

Table 12. **SfM on ETH3D [91].** Our method does not need optimization-based BA, unlike all baseline methods.

the first time an uncalibrated feed-forward approach outperforming optimization-based methods (with calibration and post-processing) on the TUM [98] dataset.

Compared to hybrid systems combining dense priors with optimization-based backends, such as MASt3R-SLAM [78] and VGGT-SLAM [69], our model achieves superior performance in the uncalibrated setting, and even surpasses MASt3R-SLAM with known calibration.

Finally, we evaluate on dynamic environments (Tab. 11), where our model again surpasses the previous SOTA uncalibrated VO, MUSt3R [10]. Neither model is trained for dynamic reconstruction, yet our raw predictions remain comparable to MegaSaM [57], a method explicitly designed for dynamic scenes and equipped with global bundle adjustment and trajectory smoothing as post-processing steps. Please refer to suppl. material/video for qualitative results.

5.8. Structure from Motion

We report SfM results in Tab. 12-13. Unlike all baselines, our method operates in a feed-forward manner without optimization-based BA. Despite this, it achieves substantial improvements in rotation accuracy over MASt3R-SfM [24] on ETH3D [91], an unordered image collection. On Tanks&Temples [50, 89] with video input, our method further surpasses ACE-Zero [8] by a large margin. Please see suppl. material/video for extensive qualitative results.

Method	**Training** (7 scenes)		**Intermediate** (8 scenes)		**Advanced** (6 scenes)	
	RRA@5	RTA@5	RRA@5	RTA@5	RRA@5	RTA@5
ACE-Zero [8]	73.9	72.9	67.6	74.0	22.9	19.1
MASt3R-SfM [24]	56.2	64.9	50.8	57.5	38.8	36.5
AMB3R (SfM)	95.0	94.5	98.7	96.9	68.0	72.4

Table 13. **SfM on Tanks&Temples [50].** Each scene contains 151 to 1106 images. We only retain methods succeed on every scene.

Method	ETH3D			DTU			7-Scenes		
	rel↓	Acc↓	Cp↓	rel↓	Acc↓	Cp↓	rel↓	Acc↓	Cp↓
w/o backend	6.02	12.81	11.89	0.83	**0.22**	0.08	5.51	2.32	3.51
w 2D backend	5.32	11.78	12.78	0.92	0.23	0.19	5.15	1.92	3.10
w/o $(\mathcal{S}^{-1} \circ f_\theta \circ \mathcal{S})$	**4.47**	10.38	11.37	1.02	0.22	0.21	5.04	2.29	3.16
Full	4.64	**9.98**	**9.69**	**0.81**	0.22	**0.08**	**4.70**	**1.75**	**2.22**

Table 14. **Ablation study on backend.** We ablate on our 3D backend versus 2D backend implemented by alternating attention.

w/o backend (VGGT)	w/o keyframe management	w/o re-sample first frame	w/o backward search (loop)	w/o max pose threshold	Full
3.6	OOM	3.7	3.8	3.4	**3.1**

Table 15. **Ablation study on AMB3R-VO** on TUM dataset.

w/o backend		w/o sub-cluster		w/o global map		Full	
RRA@5	RTA@5	RRA@5	RTA@5	RRA@5	RTA@5	RRA@5	RTA@5
93.5	73.5	97.3	73.7	94.9	74.1	**98.2**	**81.9**

Table 16. **Ablation study on AMB3R-SfM** on ETH3D dataset.

6. Discussion

6.1. Analysis

Ablation on Backend. Tab. 14 compares full model with 1) w/o backend 2) 2D backend with alternating attention and 3) w/o$(\mathcal{S}^{-1} \circ f_\theta \circ \mathcal{S})$. Our 3D backend consistently outperforms the 2D one, highlighting the advantage of maintaining a sparse yet compact 3D scene representation for explicit geometric reasoning.

Ablation on VO/SfM. We ablate different components of AMB3R-VO and AMB3R-SfM in Tab. 15 and 16. Notably, AMB3R-VO with VGGT as base model outperforms VGGT-SLAM [69] (3.6 vs 5.3), proving the superiority of the AMB3R-VO pipeline by effectively integrating the powerful prior in pointmap models that is overlooked by prior optimization-based approaches.

6.2. Conclusion

We presented AMB3R, a feed-forward model that integrates a sparse yet compact volumetric scene representation as its backend. Our paper shows this spatial compactness leads to state-of-the-art performance across diverse 3D vision tasks—including monocular/multi-view metric-scale depth/3D reconstruction, uncalibrated visual odometry, and structure-from-motion. Notably, all these results are achieved without test time optimization or task-specific fine-tuning. This shows a step towards a scalable, unified, and generalizable feed-forward 3D perception system.

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